Aaradhya Bapna

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Education

University of Waterloo Sep 2025 – Present
Honours Mathematics and Computer Science *Waterloo, ON*

- Relevant Coursework: Data Types and Structures, Calculus, Probability, Linear Algebra

Experience

Waterloo Rocketry Oct 2026 – Present
Software Developer *Waterloo, ON*

- Developing features for Omnibus, the team's real-time data acquisition dashboard, which streams sensor data from **6** channels at **40 Hz** during engine and launch testing
- Implementing Socket.IO integration to visualize high-frequency sensor data in the browser, eliminating the need for Python clients and enabling simultaneous remote viewers during critical engine tests

Projects

Waterloot | *React, Next.js, TypeScript, Supabase, PostgreSQL, Tailwind CSS* | waterloot.ca Jan 2026 – Mar 2026

- Co-built and shipped a peer-to-peer marketplace for UW students at **DeltaHacks 12**, reaching **50+ active users** and **20+ live listings** in its first week
- Built campus-only access via **@uwaterloo.ca** email verification and Supabase RLS policies so every seller is a verified student, addressing the trust problems of platforms like Facebook Marketplace
- Implemented an admin dashboard for moderation, letting admins ban users, remove listings, and resolve reports to keep the platform clean and safe

WatAsk | *Python, FastAPI, JavaScript, Supabase pgvector, Gemini API* | watask.onrender.com May 2026 – Jul 2026

- Built and shipped a full-stack application that helps students plan their degrees, covering **18** majors and double-degree programs, deployed publicly and iterated on structured feedback from over **20** students
- Built a RAG system over 1,600+ courses with Gemini semantic search and pgvector, tuned for high-precision retrieval of relevant course reviews
- Built a course dependency graph that resolves complex prerequisite chains into valid schedules, validated against official UW degree checklists

LLM Inference Engine | *Python, PyTorch, FastAPI, Docker* Oct 2026 – Present

- Building an optimized inference server achieving **5x** higher throughput than naive generation through KV caching and request batching on consumer GPUs
- Reducing time-to-first-token by **70%** while maintaining OpenAI-compatible API for drop-in integration into production applications
- Validating performance improvements against standard baselines and industry implementations through comprehensive benchmarking across latency, throughput, and concurrency metrics

Hospital Management System | *Python, SQL* Oct 2025 – Nov 2025

- Built a records system providing full CRUD operations over patients, doctors, and appointments, replacing a manual spreadsheet workflow for school-lab staff
- Designed the relational schema and indexed queries so lookups stayed fast as record counts grew, with constraints preventing double-booked appointment slots
- Separated data access from application logic behind a dedicated query layer, using parameterized SQL statements for every read and write to keep call sites free of string-built queries

Technical Skills

Languages: Python, TypeScript, JavaScript, SQL, HTML/CSS, Racket
Frameworks & Tools: React, Next.js, FastAPI, PyTorch, Hugging Face Transformers, Supabase, PostgreSQL, MySQL
AI & APIs: KV Caching, RAG, Supabase pgvector
Tools: Git, Docker, VS Code, Vercel, Excel
Libraries: Tailwind CSS, Zod, Socket.IO

Extracurriculars

Treasurer | *Movie Watchers Club* Jan 2026 – Apr 2026

- Managed a $200 CAD term budget for a **20+** member club, tracking weekly spend on supplies and events so no movie night ran short on essentials